

Essence of Linear Algebra - Full Course Notes

Based on the series by 3Blue1Brown

Chapter 1: Vectors

This video from **3Blue1Brown** introduces the fundamental building block of linear algebra: the **Vector**. It explains vectors through three different lenses and demonstrates the two core operations that define the field.

1. Three Perspectives on Vectors

The video categorizes how different disciplines view vectors:

- **The Physics Student:** Vectors are **arrows pointing in space**. They are defined by their length and direction. As long as these stay the same, you can move the arrow anywhere in space, and it remains the same vector.
- **The Computer Science Student:** Vectors are **ordered lists of numbers**. For example, a house might be represented as a 2D vector [area, price]. In this view, a vector is just a fancy word for a list, and its dimensionality is determined by the length of that list.
- **The Mathematician:** This view **generalizes** both. A vector can be *anything* as long as there is a sensible way to add two vectors and multiply them by a number.

2. The Linear Algebra Setting

For the purpose of linear algebra, the video suggests focusing on the **geometric arrow** perspective but with one key rule: **vectors are always rooted at the origin (0,0)**.

- **Coordinates:** A vector's coordinates are instructions. The first number tells you how far to walk along the x-axis, and the second tells you how far to walk parallel to the y-axis.
- **Notation:** To distinguish vectors from points, they are written vertically with square brackets (e.g., $\begin{bmatrix} x \\ y \end{bmatrix}$).

3. Core Operations

All of linear algebra revolves around two fundamental operations:

Vector Addition

- **Geometric Rule (Tip-to-Tail):** To add vector **A** and **B**, move **B** so its tail sits at the tip of **A**. The sum is the new vector stretching from the tail of **A** to the tip of **B**.
- **Numerical Rule:** You simply add the corresponding components of the lists together.
 - Example: $\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1+3 \\ 2-1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$.

Scalar Multiplication (Scaling)

- **Geometric Rule:** This involves stretching, squishing, or flipping a vector. The number used (like 2, 0.5, or -1.8) is called a **scalar** because its primary role is to "scale" the vector.
- **Numerical Rule:** Multiply every component in the list by that scalar.
 - Example: $2 * [3, 1] = [6, 2]$.

Chapter 2: Linear combinations, span, and basis vectors

This video continues the **Essence of Linear Algebra** series, focusing on how vectors combine to fill space. It introduces key concepts like linear combinations, span, and the formal definition of a basis.

1. Basis Vectors ($\hat{i}$ and $\hat{j}$)

While coordinates are often seen as just pairs of numbers, in linear algebra, they are viewed as **scalars** that scale special unit vectors:

- $\hat{i}$ (**i-hat**): The unit vector pointing right in the x-direction.
- $\hat{j}$ (**j-hat**): The unit vector pointing straight up in the y-direction.

When you see a vector like $[3, -2]$, think of it as scaling $\hat{i}$ by 3 and $\hat{j}$ by -2, then adding them together. These special vectors are the **basis** of our standard coordinate system.

2. Linear Combinations

A **linear combination** of two vectors $\vec{v}$ and $\vec{w}$ is any vector you can create by scaling them and adding the results together:

$$a\vec{v} + b\vec{w}$$

Where a and b are any real numbers (scalars).

3. The Span

The **span** of a set of vectors is the collection of **all possible linear combinations** you can reach using those vectors.

- **In 2D:**

- For most pairs of vectors, the span is the **entire 2D plane**.
- If the vectors point in the same direction (line up), the span is just a **single line** passing through the origin.
- **In 3D:**
 - Two vectors generally span a **flat sheet (plane)** slicing through 3D space.
 - Adding a third vector usually spans the **entire 3D space**, unless that third vector already lies on the plane spanned by the first two.

4. Linear Independence vs. Dependence

This describes whether a vector adds something new to the span:

- **Linearly Dependent:** At least one vector is "redundant." You could remove it without shrinking the span. This happens if one vector can be expressed as a linear combination of the others (it's already in their span).
- **Linearly Independent:** Each vector adds a new dimension to the span. None of the vectors can be created by combining the others.

5. Technical Definition of a Basis

With these terms defined, a **basis** for a space is formally defined as:

A set of **linearly independent** vectors that **span** that entire space.

Chapter 3: Linear transformations and matrices

This video explains the visual essence of **Linear Transformations** and their deep connection to **Matrices**. It offers a way to understand matrix-vector multiplication without relying on memorization.

1. What is a Linear Transformation?

In linear algebra, a **transformation** is essentially a function that takes a vector as input and spits out another vector as output. We call it a "transformation" to suggest **movement**—watching every point in space move to its new location.

A transformation is **linear** if it satisfies two visual properties:

1. **Lines must remain lines** (they cannot become curved).
2. **The origin must remain fixed** in place.

A helpful way to remember this is that **grid lines must remain parallel and evenly spaced** after the transformation.

2. Describing Transformations Numerically

You can describe where every point in space lands by knowing only where the two basis

vectors, $\hat{i}$ and $\hat{j}$, land.

If a vector $\vec{v}$ starts as a combination like $-1\hat{i} + 2\hat{j}$, it will end up as that **same combination** of the transformed $\hat{i}$ and $\hat{j}$.

3. Matrices as Geographic Records

A **Matrix** is just a way to package the coordinates of where the basis vectors landed:

- The **first column** is where $\hat{i}$ landed.
- The **second column** is where $\hat{j}$ landed.

Matrix-Vector Multiplication

To find where any vector $[x, y]$ lands, you take the x and y coordinates and use them as scalars for the columns of the matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} a \\ c \end{bmatrix} + y \begin{bmatrix} b \\ d \end{bmatrix}$$

This shows that matrix-vector multiplication is just a way to compute a **linear combination of the transformed basis vectors**.

4. Common Transformation Examples

- **90° Rotation:** $\hat{i}$ moves to $[0, 1]$ and $\hat{j}$ moves to $[-1, 0]$. The matrix is $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.
- **Shear:** $\hat{i}$ stays at $[1, 0]$ but $\hat{j}$ moves to $[1, 1]$. The matrix is $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$.
- **Linear Dependence:** If the transformed $\hat{i}$ and $\hat{j}$ land on the same line, the entire 2D plane is "squashed" down into a 1D line.

Chapter 4: Matrix multiplication as composition

This video explains the geometric intuition behind **Matrix Multiplication**, framing it as the **composition** of two or more linear transformations.

1. Composition of Transformations

Oftentimes, you want to describe the effect of applying one transformation and then another (e.g., a rotation followed by a shear). The overall effect is a new linear transformation called the **composition**.

- **Geometric View:** If you first apply a rotation and then a shear, the composition is a single matrix that captures the final landing spots of the basis vectors $\hat{i}$ and $\hat{j}$ after both actions are completed.
- **Notation:** Matrix multiplication is read from **right to left**, mimicking function notation $f(g(x))$. If you apply transformation A and then B , the resulting matrix is written as BA .

2. Numerical Matrix Multiplication

To find the product of two matrices without an animation, you follow the basis vectors:

1. Take the **first column** of the right matrix (where $\hat{i}$ initially lands).
2. Multiply it by the left matrix to see where it ends up after the second transformation. This result is the **first column** of the new composition matrix.
3. Repeat the process for the **second column** (where $\hat{j}$ lands).

3. Key Properties

Thinking of multiplication as composition makes its properties much more intuitive:

- **Order Matters (Non-commutative):** Applying a shear then a rotation is usually different from applying a rotation then a shear ($AB \neq BA$). You can visualize this by seeing how different the final space looks depending on the order of steps.
- **Associativity:** $(AB)C = A(BC)$. This states that applying C , then B , then A is the same regardless of how you group them. Geometrically, this is trivial because the sequence of transformations remains the same.

Chapter 5: Three-dimensional linear transformations

This video explains how the concepts of linear transformations and matrices extend from two dimensions into **three-dimensional space**. The core logic remains the same, but with an additional axis and basis vector.

1. Visualizing 3D Transformations

In 3D, a linear transformation "smushes" space while keeping grid lines parallel and evenly spaced and fixing the origin in place.

To describe these movements without getting overwhelmed by the full 3D grid, we focus on where the **three basis vectors** land:

- $\hat{i}$ (**i-hat**): Unit vector in the x-direction.
- $\hat{j}$ (**j-hat**): Unit vector in the y-direction.
- $\hat{k}$ (**k-hat**): Unit vector in the z-direction.

2. 3x3 Matrices

A 3D linear transformation is completely described by a **3x3 matrix**. Each column represents the final coordinates of one of the basis vectors:

- **Column 1:** Where $\hat{i}$ lands.
- **Column 2:** Where $\hat{j}$ lands.
- **Column 3:** Where $\hat{k}$ lands.

Example: 90° Rotation around the Y-axis

- $\hat{i}$ moves to the negative z-axis: $[0, 0, -1]$
- $\hat{j}$ stays fixed: $[0, 1, 0]$
- $\hat{k}$ moves to the x-axis: $[1, 0, 0]$

- **Matrix:**
$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

3. Matrix-Vector Multiplication

To find where a 3D vector $[x, y, z]$ lands after a transformation, you use the coordinates as scalars for the columns of the matrix:

$$\text{Output} = x(\text{Col 1}) + y(\text{Col 2}) + z(\text{Col 3})$$

This is the same "scaling and adding" process used in 2D, just with one extra step for the z-component.

4. 3D Matrix Composition

Multiplying two 3x3 matrices represents applying one 3D transformation after another. This is crucial in fields like **computer graphics and robotics**, where complex 3D rotations are often broken down into a sequence of simpler, easier-to-calculate rotations.

Chapter 6: The determinant

This video from **3Blue1Brown** explains the **Determinant**, focusing on its geometric meaning as a scaling factor for area and volume.

1. The Core Idea: Scaling Factor

A linear transformation might stretch or squish space. The **determinant** is a single number that measures exactly by how much the transformation scales areas (in 2D) or volumes (in 3D).

- **Determinant = 2:** Every region in space now has twice its original area.
- **Determinant = 1:** Areas remain the same, even if the shape is warped (like in a shear).
- **Determinant = 0:** Space is "squashed" into a lower dimension (e.g., a 2D plane is flattened into a line). This means the columns of the matrix are **linearly dependent**.

2. Negative Determinants and Orientation

A determinant can be negative, which relates to **orientation**:

- **In 2D:** A negative determinant means space has been **flipped over** (like turning over a sheet of paper). You can tell this happened if $\hat{j}$ was originally to the left of $\hat{i}$, but ends up on the right after the transformation.
- **In 3D:** You can use the **Right-Hand Rule**. If the transformation forces you to use your left hand to match the positions of $\hat{i}$, $\hat{j}$, and $\hat{k}$, the orientation has flipped, and the determinant is negative.

3. Higher Dimensions (3D)

In three dimensions, the determinant measures the scaling factor for **volume**.

- Imagine the $1 \times 1 \times 1$ cube formed by the basis vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.
- The transformation warps this cube into a slanted shape called a **parallelepiped**.
- The volume of this parallelepiped is the determinant of the 3x3 matrix.

4. Computing the Determinant

For a 2x2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the formula is:

$$\det(A) = ad - bc$$

The ad term represents the area of a rectangle if the off-diagonal terms b and c were zero. The bc term accounts for how much the shape is stretched or squished diagonally.

5. Why $\det(AB) = \det(A)\det(B)$?

Thinking of the determinant as a scaling factor makes this property intuitive:

If you scale area by factor B and then scale that new area by factor A , the total scaling factor is simply the product $A \times B$.

Chapter 7: Inverse matrices, column space and null space

This video from **3Blue1Brown** explains how linear algebra helps solve systems of equations by viewing matrices as transformations. It covers inverse matrices, column space, rank, and null space.

1. Solving Systems of Equations Geometrically

A linear system of equations can be written as $Ax = v$, where A is a matrix, x is a vector of unknowns, and v is a constant vector.

- Geometrically, this means we are looking for a vector x that, after being transformed by A , lands exactly on v .

2. Inverse Matrices (A^{-1})

If the determinant of A is **non-zero**, space does not get squished. In this case, there is a unique transformation that "undoes" the original one, called the **inverse matrix**.

- Applying A and then A^{-1} brings you back to where you started, which is the **identity transformation** (I).
- To solve for x , you "play the tape in reverse" by multiplying the inverse by v : $x = A^{-1}v$.

3. Column Space and Rank

When the determinant is **zero**, space is squished into a lower dimension, and no unique inverse exists.

- **Column Space:** The set of all possible outputs of a transformation (where all input vectors land). It is the **span** of the columns of the matrix.
- **Rank:** The number of dimensions in the column space (e.g., if the output is a line, the rank is 1; if it's a plane, the rank is 2).
- **Full Rank:** When the rank equals the number of columns, meaning space is not squished into a lower dimension.

4. Null Space (Kernel)

For transformations that squish space, many different input vectors might land on the same output point.

- **Null Space:** The set of all vectors that land exactly on the **origin (zero vector)** after the transformation.
- If you are solving $Ax = 0$, the null space represents all possible solutions to that equation.

Chapter 8: Nonsquare matrices as transformations between dimensions

This video explains the geometric meaning of **non-square matrices** as linear transformations that map vectors from one dimension to another.

1. Transformations Between Dimensions

While square matrices (like 2x2 or 3x3) transform space within the same dimension, non-square matrices represent mappings between *different* spaces.

- **Linearity remains the same:** Grid lines must stay parallel and evenly spaced, and the origin must remain fixed.
- **Columns still represent basis vectors:** The number of columns tells you the dimension of the **input space**, while the number of rows (the number of coordinates in each column) tells you the dimension of the **output space**.

2. Mapping 2D to 3D (3x2 Matrix)

A **3x2 matrix** (3 rows, 2 columns) takes 2D vector inputs and spits out 3D vector outputs.

- **Columns:** There are 2 columns because the input space has 2 basis vectors ($\hat{i}$ and $\hat{j}$).
- **Rows:** Each column has 3 entries because the landing spot for those basis vectors is in 3D space.

- **Geometry:** The output (column space) is typically a **2D plane** slicing through 3D space.

3. Mapping 3D to 2D (2x3 Matrix)

A **2x3 matrix** (2 rows, 3 columns) maps 3D inputs to 2D outputs.

- **Columns:** 3 columns represent the 3 original basis vectors $(\hat{i}, \hat{j}, \hat{k})$.
- **Rows:** 2 rows mean each basis vector lands in a 2D plane.
- **Geometry:** This is a "squashing" transformation where a 3D volume is flattened onto a 2D surface.

4. Mapping 2D to 1D (1x2 Matrix)

A **1x2 matrix** takes in 2D vectors and outputs a single number (a point on the number line).

- This transformation is particularly important because it is closely related to the **dot product**, which will be the focus of the next video in the series.

Chapter 9: Dot products and duality

This video explores the **Dot Product** through the lens of linear transformations and a concept called **Duality**, explaining why the numerical calculation relates so perfectly to geometric projection.

1. The Standard View

Numerically, the dot product of two vectors is the sum of the products of their corresponding coordinates.

- **Geometric Interpretation:** Project vector **W** onto the line passing through vector **V**, then multiply the length of that projection by the length of **V**.
 - **Positive:** Vectors point in generally the same direction.
 - **Zero:** Vectors are perpendicular.
 - **Negative:** Vectors point in generally opposite directions.

2. Linear Transformations to 1D

To understand *why* the dot product works, we look at linear transformations that map a 2D vector to a 1D number line.

- **Visual Test:** Any line of evenly spaced dots in the input space must remain evenly spaced on the output number line.
- **Matrix Representation:** These are 1×2 matrices. For example, if $\hat{i}$ lands on 1 and $\hat{j}$ lands on -2, the matrix is $[1, -2]$.

3. The Dot Product/Projection Connection

Imagine a diagonal number line in 2D space and a unit vector $\hat{u}$ pointing to "1" on that line. Projecting space onto this line defines a linear transformation.

- **Symmetry Argument:** By symmetry, the number where $\hat{i}$ lands when projected onto the line is the same as the x-coordinate of $\hat{u}$.
- **Result:** The 1×2 matrix for this projection transformation is just the coordinates of the vector you are projecting onto.

4. Duality

This reveals a deep mathematical link called **Duality**. It is the correspondence between two different types of mathematical objects:

- **Vector:** An arrow in space.
- **Dual Transformation:** A linear transformation that squashes space onto a number line.

Applying a transformation to a vector is computationally identical to taking the dot product of that vector with the "dual" of the transformation.

Chapter 10: Cross products

This video introduces the **Cross Product**, first showing its behavior in two dimensions before explaining the true three-dimensional operation and its geometric significance.

1. The 2D "Cross Product" (Area and Orientation)

In two dimensions, the cross product of vectors **V** and **W** is essentially the signed area of the parallelogram they span.

- **Magnitude:** The area of the parallelogram formed by **V** and **W**.
- **Orientation:**
 - If **V** is to the right of **W**, the value is **positive**.
 - If **V** is to the left of **W**, the value is **negative**.
- **Calculation:** You write the coordinates of **V** and **W** as the columns of a matrix and compute the **determinant**. This works because the determinant measures how the area of the unit square changes under a transformation.

2. The 3D Cross Product

The true cross product takes two 3D vectors and produces a **new 3D vector**.

- **Length (Magnitude):** The length of the resulting vector equals the area of the parallelogram defined by **V** and **W**.
- **Direction:** The resulting vector is **perpendicular** to the plane containing **V** and **W**.
- **The Right-Hand Rule:** To determine which of the two perpendicular directions the vector points, use your right hand:

1. Point your index finger in the direction of $\mathbf{V}$.
2. Point your middle finger in the direction of $\mathbf{W}$.
3. Your thumb points in the direction of $\mathbf{V} \times \mathbf{W}$.

3. Numerical Computation

The standard way to calculate the cross product involves a specific determinant process that looks a bit strange at first:

1. Create a 3x3 matrix.
2. The **first row** contains the basis vectors $\hat{i}$, $\hat{j}$, and $\hat{k}$.
3. The **second row** contains the coordinates of $\mathbf{V}$.
4. The **third row** contains the coordinates of $\mathbf{W}$.
5. Compute the determinant. The result is a linear combination of $\hat{i}$, $\hat{j}$, and $\hat{k}$, which forms your new vector.

Chapter 11: Cross products in the light of linear transformations

This video provides an elegant explanation for the **3D Cross Product** by connecting the strange computational "trick" (using $\hat{i}$, $\hat{j}$, $\hat{k}$ in a matrix) to the underlying geometry through the lens of **Linear Transformations** and **Duality**.

1. The Setup: A Linear Transformation to 1D

To understand the cross product $\mathbf{V} \times \mathbf{W}$, we define a special linear transformation $f(u)$ that takes any 3D vector $u = [x, y, z]$ and returns a number:

- **The Function:** $f(u) = \det([u, \mathbf{V}, \mathbf{W}])$.
- **Geometric Meaning:** This function calculates the **signed volume** of the parallelepiped formed by u and the two fixed vectors $\mathbf{V}$ and $\mathbf{W}$.
- **Linearity:** Because the determinant is linear with respect to its first column, $f(u)$ is a linear transformation from 3D space to the 1D number line.

2. Duality: Finding the Dual Vector

The principle of **Duality** states that every linear transformation from a space to the number line can be represented as a **dot product** with a unique "dual vector" $\mathbf{P}$:

$$f(u) = P \cdot u$$

We want to find this vector P .

The Computational Approach

If you calculate the determinant $\det([u, V, W])$ algebraically, you get a linear combination of x , y , and z . The coefficients of these variables are the coordinates of our dual vector P .

This is exactly what the notational trick (putting $\hat{i}, \hat{j}, \hat{k}$ in the first row) is doing—it's a shortcut to find these coordinates.

The Geometric Approach

We can also find P by looking at the geometry of the volume calculation:

- **Volume formula:** $\text{Volume} = (\text{Area of base}) \times (\text{perpendicular height})$.
- The base is the parallelogram formed by V and W . Let its area be A .
- The perpendicular height is the projection of u onto a line perpendicular to V and W .
- This matches the geometric definition of a dot product $P \cdot u$, where P is a vector perpendicular to V and W with a length equal to the area A .

3. Conclusion: The Two Views Are the Same

Because both the "weird" matrix calculation and the geometric volume projection describe the same linear transformation, they must be identifying the **same vector**.

- **The Matrix Calculation** gives us the coordinates.
- **The Geometry** gives us the perpendicularity and the area-length relationship.

Chapter 12: Cramer's rule, explained geometrically

This video explains the geometric intuition behind **Cramer's Rule**, a method for solving linear systems of equations. While not the most efficient for large systems, it offers a beautiful connection between determinants and coordinates.

1. The Core Idea: Coordinates as Areas

Normally, we think of a vector's coordinates (x, y) as its distance along the axes. However, we can also think of them geometrically:

- The y -coordinate is the area of a parallelogram formed by the basis vector $\hat{i}$ and the vector (x, y) .
- The x -coordinate is the area of a parallelogram formed by the vector (x, y) and the basis vector $\hat{j}$.

In 3D, the z -coordinate is the volume of a parallelepiped formed by the basis vectors $\hat{i}, \hat{j}$, and the mystery vector.

2. Transformations and Area Scaling

The key insight of linear algebra is that every linear transformation scales all areas by a single factor: the **determinant**.

If we have a linear system $Ax = v$:

1. The vector x is our "mystery input."
2. The vector v is the "known output."
3. The parallelogram formed by $\hat{i}$ and x has an area of y .
4. After the transformation A , this parallelogram is now formed by the **first column of A** and the **output vector** v .

Because all areas were scaled by $\det(A)$, the area of this new parallelogram is:

$$\text{Area}_{\text{new}} = y \times \det(A)$$

3. Solving for y (and x)

We can solve for y by rearranging the formula:

$$y = \frac{\text{Area}_{\text{new}}}{\det(A)}$$

The new area is simply the determinant of a matrix where the second column (originally $\hat{j}$) has been replaced by the output vector v . Similarly, to find x , you replace the first column of the matrix with v .

4. Cramer's Rule Formula

For a 2x2 system, the solutions are:

$$x = \frac{\det \begin{bmatrix} v_1 & a_{12} \\ v_2 & a_{22} \end{bmatrix}}{\det(A)}, \quad y = \frac{\det \begin{bmatrix} a_{11} & v_1 \\ a_{21} & v_2 \end{bmatrix}}{\det(A)}$$

Chapter 13: Change of basis

This video from the **Essence of Linear Algebra** series explains the concept of **Change of Basis**, which is essentially a way to translate between different "languages" used to describe vectors in space.

1. What is a Basis?

Every coordinate system relies on a set of **basis vectors**. In our standard system, these are $\hat{i}$ (right, length 1) and $\hat{j}$ (up, length 1).

- A vector like $[3, 2]$ is a shorthand for $3\hat{i} + 2\hat{j}$.
- If someone else, say Jennifer, uses different basis vectors (let's call them $\vec{b}_1$ and $\vec{b}_2$), she would use different coordinates to describe the exact same arrow in space.

2. Translating Vectors Between Systems

To translate from Jennifer's "language" to ours, we use a **Change of Basis Matrix**.

- **The Matrix:** You create a matrix where the columns are Jennifer's basis vectors ($\vec{b}_1$ and

$\vec{b}_2$) written in our coordinates.

- **Translating Jennifer to Us:** Multiply this matrix by her coordinates for a vector to find its coordinates in our system.
- **Translating Us to Jennifer:** Multiply the **inverse** of that matrix by our coordinates to find how Jennifer would describe the vector.

3. Translating Transformations

The most powerful use of change of basis is describing a transformation (like a rotation) in another system.

- To see a transformation M through Jennifer's eyes, you use the formula:

$$P^{-1}MP$$

Where P is the change of basis matrix.

This process involves three steps:

1. P : Translate Jennifer's coordinates into our coordinates.
2. M : Apply the transformation in our "familiar" language.
3. P^{-1} : Translate the result back into Jennifer's language.

Summary

- **Matrices** can be thought of as a record of where basis vectors land.
- **Change of basis** is a "shift in perspective."
- The expression $P^{-1}MP$ represents "the same transformation, but as someone else sees it." This concept is vital for understanding more advanced topics like eigenvalues and diagonalizing matrices.

Chapter 14: Eigenvectors and eigenvalues

This video from **3Blue1Brown** explains the geometric intuition behind **Eigenvectors** and **Eigenvalues**, showing how they describe the "purity" of a transformation without relying on specific coordinate systems.

1. What are Eigenvectors and Eigenvalues?

Most vectors are knocked off their span (the line passing through the origin and the vector's tip) during a linear transformation.

- **Eigenvectors:** These are special vectors that **remain on their own span** during a

transformation. The only thing that happens to them is that they are stretched or squished.

- **Eigenvalues:** This is the **factor** by which an eigenvector is stretched or squished. For example, an eigenvalue of 3 means the vector becomes three times longer.

2. Physical Examples

- **3D Rotation:** If you find an eigenvector for a 3D rotation, you have found the **axis of rotation** (since vectors on that axis stay on their span). The eigenvalue for this axis is always 1.

3. The Characteristic Equation

To find eigenvalues (λ) and eigenvectors (v) for a matrix A , we solve the equation:

$$Av = \lambda v$$

This can be rewritten as:

$$(A - \lambda I)v = 0$$

For a non-zero vector v to satisfy this, the matrix $(A - \lambda I)$ must squish space into a lower dimension, which means its **determinant must be zero**:

$$\det(A - \lambda I) = 0$$

4. Special Cases

- **Rotations (2D):** A 90° rotation has no real eigenvectors because every vector is moved off its span. Computationally, this results in imaginary eigenvalues.
- **Shears:** A shear transformation only has one line of eigenvectors (the fixed axis) with an eigenvalue of 1.

5. Eigenbasis and Diagonalization

If a transformation has enough eigenvectors to span the entire space, you can choose them as

your basis vectors. This is called an **Eigenbasis**.

- **Diagonal Matrices:** In an eigenbasis, the transformation matrix becomes **diagonal**, with eigenvalues sitting on the main diagonal and zeros everywhere else.
- **Why it matters:** Diagonal matrices are incredibly easy to work with. For example, computing the 100th power of a diagonal matrix just requires raising each diagonal entry to the 100th power.

Chapter 15: A quick trick for computing eigenvalues

This video from **3Blue1Brown** presents a much faster method for computing **eigenvalues** of 2x2 matrices without setting up the full characteristic polynomial. It relies on the relationship between a matrix's trace, its determinant, and its eigenvalues.

1. The Core Relationships

For any 2x2 matrix, there are two key properties that link its entries to its eigenvalues (λ_1 and λ_2):

1. **Trace (Mean Property):** The sum of the diagonal entries (the trace) is equal to the sum of the eigenvalues. More helpfully, the **mean of the diagonal entries** (m) is equal to the **mean of the eigenvalues**.
2. **Determinant (Product Property):** The determinant of the matrix ($ad - bc$) is equal to the **product of the eigenvalues** (p).

2. The "Mean-Product" Formula

Once you know the mean (m) and product (p) of two numbers, you can find the numbers themselves using this formula:

$$\lambda = m \pm \sqrt{m^2 - p}$$

This is essentially a more intuitive rephrasing of the quadratic formula.

3. Step-by-Step Examples

Example 1: Matrix $\begin{bmatrix} 3 & 1 \\ 4 & 1 \end{bmatrix}$

- **Mean (m):** Average of the diagonals 3 and 1 is **2**.
- **Product (p):** Determinant is $(3 \times 1) - (1 \times 4) = -1$.
- **Eigenvalues:** $2 \pm \sqrt{2^2 - (-1)} = 2 \pm \sqrt{5}$.

Example 4: Matrix $\begin{bmatrix} 2 & 1 \\ 7 & 8 \end{bmatrix}$

- **Mean (m):** Average of 2 and 8 is **5**.
- **Product (p):** Determinant is $(2 \times 8) - (7 \times 1) = 9$.
- **Eigenvalues:** $5 \pm \sqrt{5^2 - 9} = 5 \pm \sqrt{16} = 9, 1$.

4. Real-World Application: Quantum Mechanics

This trick is particularly useful for the **Pauli spin matrices** used in physics. These matrices all have a trace of 0 (mean = 0) and a determinant of -1 (product = -1), which leads directly to eigenvalues of **+1 and -1**.

Chapter 16: Abstract vector spaces

This final video in the **Essence of Linear Algebra** series explores **Abstract Vector Spaces**, moving beyond arrows and lists of numbers to show how the rules of linear algebra apply to many other mathematical objects, most notably functions.

1. What are Vectors, Really?

The series began by asking if vectors are arrows or lists of numbers. The mathematician's answer is more general: a vector is **anything** that follows a specific set of rules for **addition** and **scalar multiplication**.

- **Example: Functions as Vectors.** You can add two functions together (by adding their outputs) and you can scale a function by a number. Because they follow these rules, functions can be treated as vectors.
 - Adding $f(x)$ and $g(x)$ is similar to adding coordinates, except there are **infinitely many coordinates** (one for every possible value of x).

2. Linear Transformations in the Abstract

A transformation is **linear** if it satisfies two properties, regardless of whether it acts on arrows or functions:

- **Additivity:** $L(\vec{v} + \vec{w}) = L(\vec{v}) + L(\vec{w})$

- **Scaling:** $L(c\vec{v}) = cL(\vec{v})$

The Derivative as a Linear Transformation: In calculus, the derivative operator is a linear transformation. If you add two functions and then take the derivative, it's the same as taking the derivatives separately and adding them.

3. Representing Derivatives with Matrices

It is possible to represent the derivative of a polynomial with a matrix by choosing a basis (like $1, x, x^2, x^3, \dots$).

- In this coordinate system, a polynomial like $x^3 + 5x^2 + 4x + 5$ is a vector of its coefficients: $[5, 4, 5, 1, 0, \dots]$.
- The derivative operation is then represented by an **infinite matrix**. Multiplying the matrix by the vector gives the coefficients of the derivative function.

4. Vector Spaces and Axioms

A **Vector Space** is any collection of objects that satisfy **eight axioms** regarding addition and scaling.

- These axioms act as a "checklist." If your objects (arrows, functions, or even "pi creatures") pass the test, every discovery in linear algebra—eigenvectors, determinants, inverses—automatically applies to them.

Summary

- **Linear Algebra** is not just about arrows; it is a general framework for any system that follows linear rules.
- **Abstracting** vectors into vector spaces allows mathematicians to prove results once and apply them to countless different fields, from quantum mechanics to computer science.
- Understanding the **geometric "essence"** (like grid lines and transformations) is the best way to build a foundation for this abstract power.